SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SMART STUDENT HUB

Project Title: Smart Student Hub: Task Tracking & Deadline Reminder System for UNESWA students

Prepared By:

- **Tamile Ntshalintshali – Team Leader**
- **Nosipho Shabangu – System Architect and Design**
- **Karabo Nkonyane – Programmer**

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Smart Student Hub system. The document is intended for developers, supervisors, lecturers, testers, and project stakeholders involved in the development of the system.The purpose of this document is to clearly define the system requirements needed to develop the Smart Student Hub successfully.

1.2 Scope

The Smart Student Hub is a web-based task tracking and deadline reminder system developed for students at the University of Eswatini. The system allows students to create academic task, track assignment deadlines, receive reminder notifications and manage pending and completed academic activities. The system aims to improve academic organization, time management, and productivity among university students.

The system will not include:

- GPA calculations
- Online examinations
- Financial management services

1.3 Definitions, Acronyms, and Abbreviations

Term	**Meaning**
SRS	Software Requirements Specification
UI	User Interface
DBMS	Database Management System
FR	Functional Requirements
Login Authentication	Verification of user identity
Dashboard	Main interface showing tasks and reminders
Notification	Reminder alert for deadlines

1.4 References

- IEEE Std 830-1998 Software Requirements Specification Template
- Systems Analysis and Design Notes
- Smart Student Hub Project Proposal

1.5 Overview

This document contains:

- Overall system description
- Product functions
- User characteristics
- Constraints
- Functional requirements
- Performance requirements
- System attributes

The document follows the IEEE 830-1998 SRS structure.

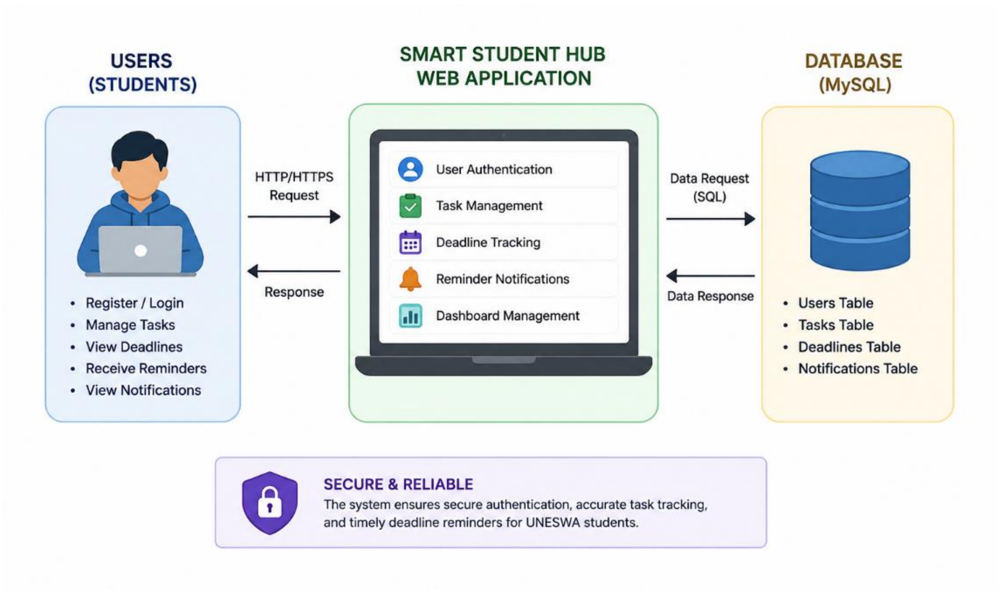
2. OVERALL DESCRIPTION

2.1 Product Perspective

The Smart Student Hub is an independent web-based academic management system designed for UNESWA students. The system helps students manage academic schedules and deadlines through:

- Task tracking
- Deadline monitoring
- Reminder notifications
- Dashboard management

<u>System Overview Diagram</u>



2.2 Product Functions

The system performs the following major functions:

a) User Authentication

- User registration
- User login
- Password verification

b) Task Management

- Add academic tasks
- Edit tasks
- Delete tasks
- Mark tasks as completed

c) Deadline Tracking

- Monitor assignment deadlines
- Display upcoming deadlines
- Show overdue tasks

d) Notification Management

- Generate reminder alerts
- Notify users before deadlines

e) Dashboard Management

- Display pending tasks
- Display completed tasks
- Display notifications

2.3 User Characteristics

Students

- Basic computer literacy
- Familiar with smartphones and web applications
- Able to manage academic activities online

2.4 Constraints

The system has the following constraints:

- Requires internet connection
- Requires modern web browser
- Depends on PHP and MySQL technologies
- Limited to academic task management only
- User authentication required for access

2.5 Assumptions and Dependencies

The system assumes:

- Students have internet access
- Users possess valid login credentials
- The server database remains operational
- The hosting environment supports PHP and MySQL

2.6 Apportioning of Requirements

The following features may be added in future versions:

- Mobile application integration
- Email and SMS reminders
- Cloud synchronization

3. SPECIFIC REQUIREMENTS

3.1 External Interface Requirement

<u>User Interface</u>

- Login screen
- Registration screen
- Dashboard screen
- Add task screen
- Notification screen

<u>The interface shall</u>:

- Be user-friendly
- Support simple navigation
- Display clear buttons and forms
- Provide feedback messages

Hardware Interface

- Desktop computers
- Laptops
- Smartphones
- Tablets

Software Interfaces

- MySQL Database Server
- PHP Web Server
- Web browsers such as Chrome and Firefox

Communications Interface

- HTTP/HTTPS communication protocols
- Internet-based client-server communication

3.2 Functional Requirements

User Authentication Module

FR1: User Registration - The system shall allow students to create accounts using: Username**,** email and password

FR2: User Login - The system shall authenticate users before granting access.

FR3: Password Validation - The system shall validate passwords before login access is granted.

Task Management Module

FR4: Add Task - The system shall allow users to add task title, description**,** deadline date and priority level

FR5: Edit Task - The system shall allow users to update task details.

FR6: Delete Task - The system shall allow users to remove tasks from the system.

FR7: Mark Task as Complete - The system shall allow users to mark completed tasks.

Deadline Reminder Module

FR8: Deadline Monitoring - The system shall monitor task deadlines continuously.

FR9: Reminder Notifications - The system shall send reminder alerts before deadlines.

FR10: Overdue Alerts - The system shall notify users about overdue tasks.

Dashboard Module

FR11: View Pending Tasks - The system shall display pending academic tasks.

FR12: View Completed Tasks - The system shall display completed tasks.

FR13: View Notifications - The system shall display reminder notifications on the dashboard.

3.3 Performance Requirements

- The system shall load dashboard information within 3 seconds.
- The system shall support multiple users simultaneously.
- Reminder notifications shall appear immediately when deadlines approach.
- User login shall complete within 5 seconds.

3.4 Design Constraints

The system shall:

- Be developed using PHP, HTML, CSS, JavaScript, and MySQL
- Operate as a web-based application
- Follow secure authentication procedures
- Support responsive web design

3.5 Software System Attribute

Reliability - The system shall save task information accurately and prevent data loss during normal operations

Availability - The system shall be available 24 hours a day and support continuous student access

Security - The system shall require login authentication, protect user passwords and restrict unauthorized access

Maintainability - The system shall support future upgrade, allow easy modification of features and use organized source code

Portability - The system shall operate on different browsers and support desktop and mobile devices

3.6 Other Requirements

The system shall provide simple navigation, display clear error messages and support easy task management for students

4. APPENDICES

Appendix A: Development Tools

- PHP
- MySQL
- HTML
- CSS
- JavaScript

Appendix B: Future Improvements

- Mobile application version
- AI-powered reminders
- Email notification integration
- Cloud storage support